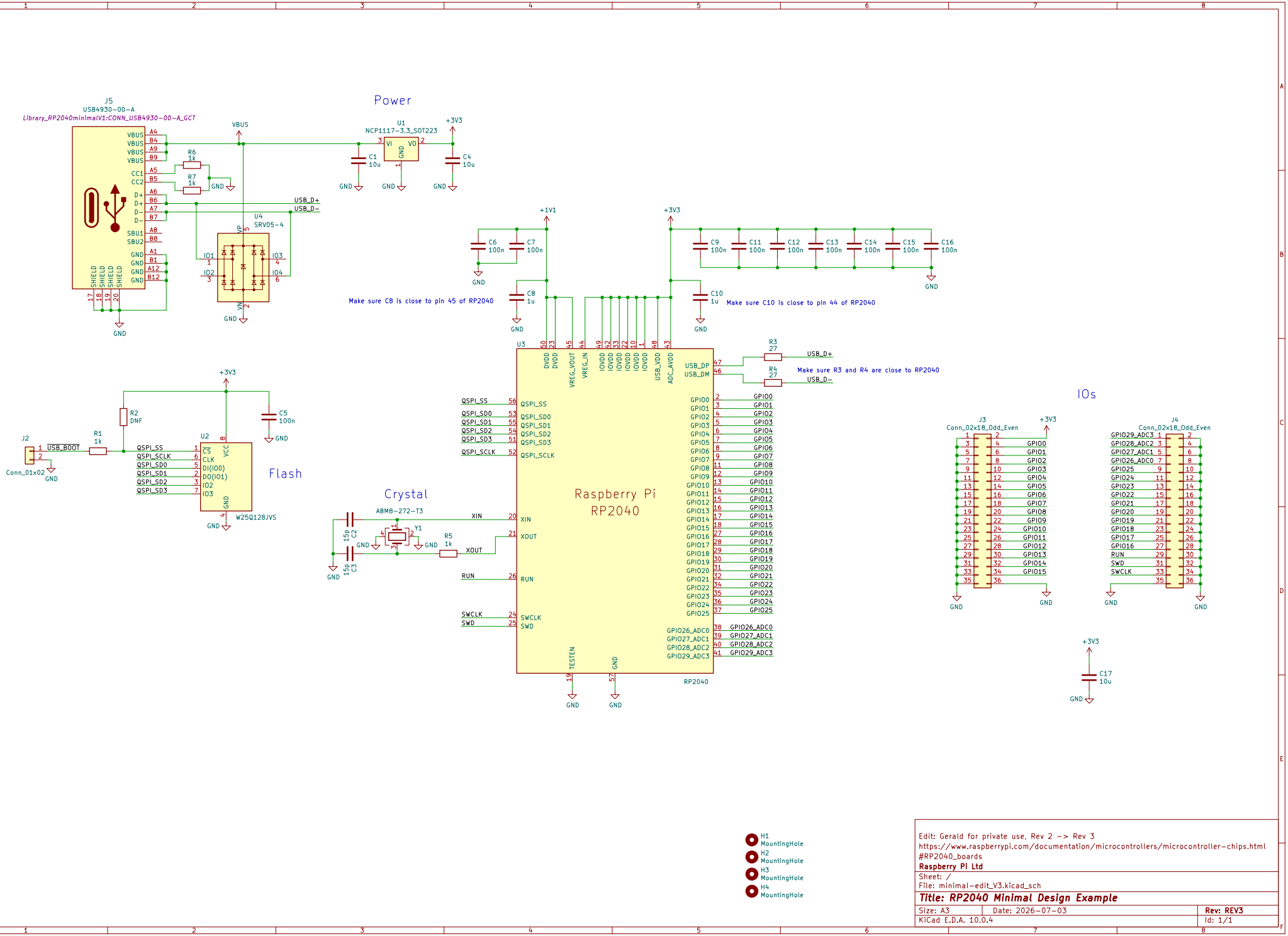

Power
Flash
Crystal
IOs
Raspberry Pi
RP2040
J5
USB4930-00-A
Library_RP2040minimalV1:CONN_USB4930-00-A_GCT
U1
NCP1117-3.3_SOT223
U2
W25Q128JVS
U3
RP2040
U4
SRV05-4
J2
Conn_01x02
J3
Conn_02x18_Odd_Even
J4
Conn_02x18_Odd_Even
Make sure C8 is close to pin 45 of RP2040
Make sure C10 is close to pin 44 of RP2040
Make sure R3 and R4 are close to RP2040
USB_D+
USB_D-
USB_BOOT
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
XIN
XOUT
RUN
SWCLK
SWD
VBUS
+3V3
+1V1
GND
Edit: Gerald for private use, Rev 2 -> Rev 3
https://www.raspberrypi.com/documentation/microcontrollers/microcontroller-chips.html#RP2040_boards
Raspberry Pi Ltd
Sheet: /
File: minimal-edit_V3.kicad_sch
Title: RP2040 Minimal Design Example
Size: A3
Date: 2026-07-03
Rev: REV3
KiCad E.D.A. 10.0.4
Id: 1/1